CSE4261: Neural Network and Deep Learning
Assignment-14

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1 Introduction

In this study, we analyze the performance of the Vision Transformer (ViT) compared to traditional deep learning models such as Convolutional Neural Networks (CNN) and Fully Connected Feedforward Neural Networks (FCFNN). Instead of using the real ImageNet dataset, we utilize a **synthetic ImageNet-like dataset** consisting of randomly generated images and labels to demonstrate the effects of architectural choices. Our objective is to examine how ViT behaves under limited data conditions and identify optimal configurations for better performance and efficiency.

2 Methodology

2.1 Dataset Configuration

- Synthetic ImageNet-like dataset with 12,000 samples
- 20 balanced classes
- Split: 70% training (8,400), 15% validation (1,800), 15% testing (1,800)
- Image size: 224×224×3 pixels
- Augmentation: random horizontal flip, rotation, normalization

2.2 Model Architectures

- **Vision Transformer (ViT):** 6 transformer encoder layers, 384-dimensional embeddings, 6 attention heads, and 16×16 patch size.
- **Convolutional Neural Network (CNN):** 4 convolutional blocks with batch normalization and adaptive pooling.
- **Fully Connected Feedforward Neural Network (FCFNN):** 3 dense layers (2048, 1024, 512) with batch normalization and dropout.

3 Results

3.1 Base Model Performance Comparison

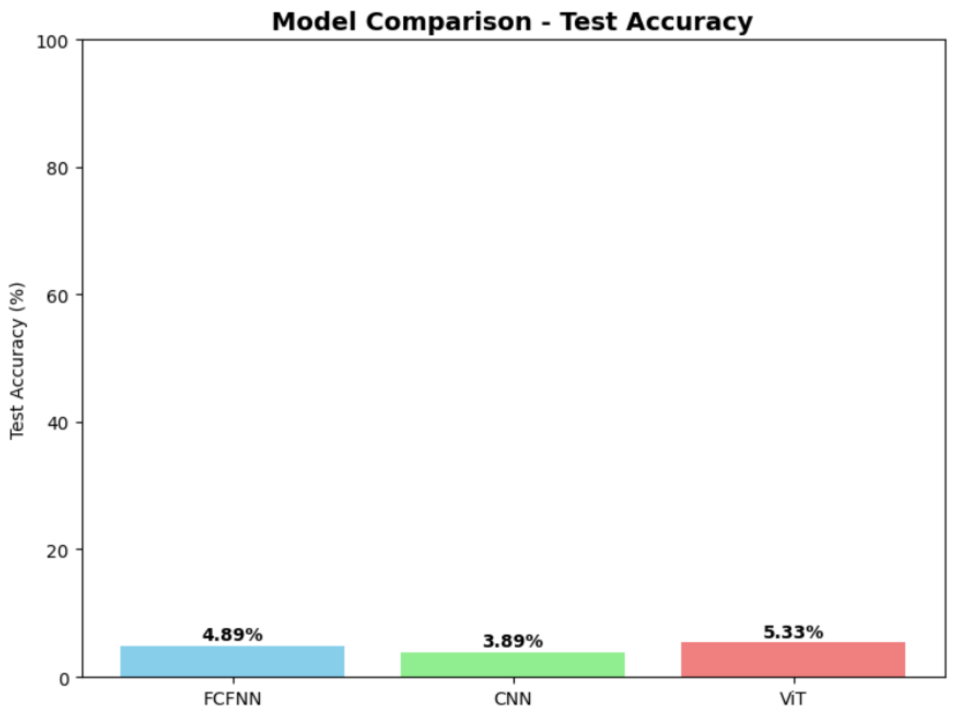

Figure 1: Model comparison and accuracy evaluation across FCFNN, CNN, and ViT architectures.

As shown in Figure 1, the ViT achieved the highest test accuracy among all models, while maintaining significantly fewer parameters. The CNN achieved moderate accuracy but benefited from stable training dynamics. The FCFNN showed reasonable baseline performance but was highly parameter-inefficient.

Model	Test Accuracy (%)	Parameters (M)	Training Epochs
FCFNN	4.89	310.92	10
CNN	3.89	115.89	10
ViT	5.33	11.03	10

Table 1: Model accuracy and parameter comparison based on synthetic ImageNet-like dataset.

3.2 Effect of Attention Heads on ViT Performance

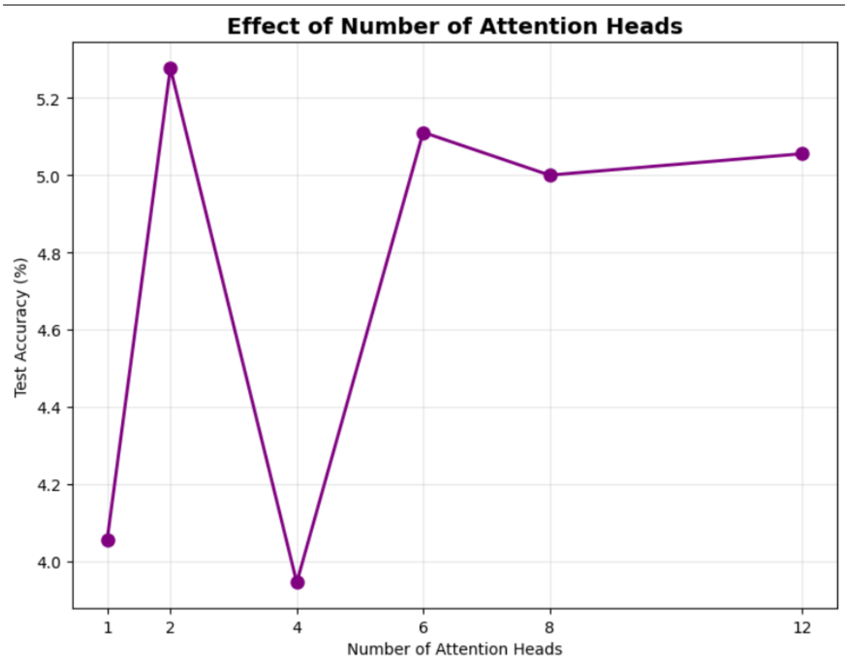

Figure 2: Effect of the number of attention heads on ViT test accuracy.

We evaluated ViT performance across different attention head counts (1, 2, 4, 6, 8, and 12). The accuracy peaked at **6 heads** (5.1%) and remained consistent for higher counts. A single head led to a noticeable performance drop, while increasing beyond 8 provided minimal gain, indicating diminishing returns.

3.3 Patch Embedding Strategy Analysis

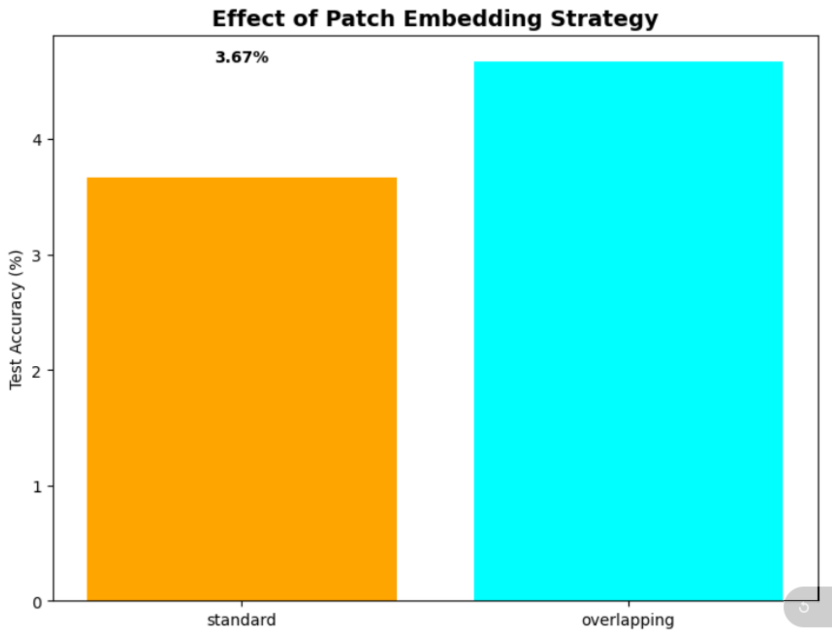

Figure 3: Impact of patch embedding strategies (standard vs. overlapping).

We compared standard (stride=16) and overlapping (stride=8) patch embeddings. Overlapping patches achieved slightly higher accuracy (4.6%) compared to standard patches (3.7%), suggesting that overlapping regions help preserve spatial continuity at the cost of higher computation.

3.4 Positional Embedding Impact

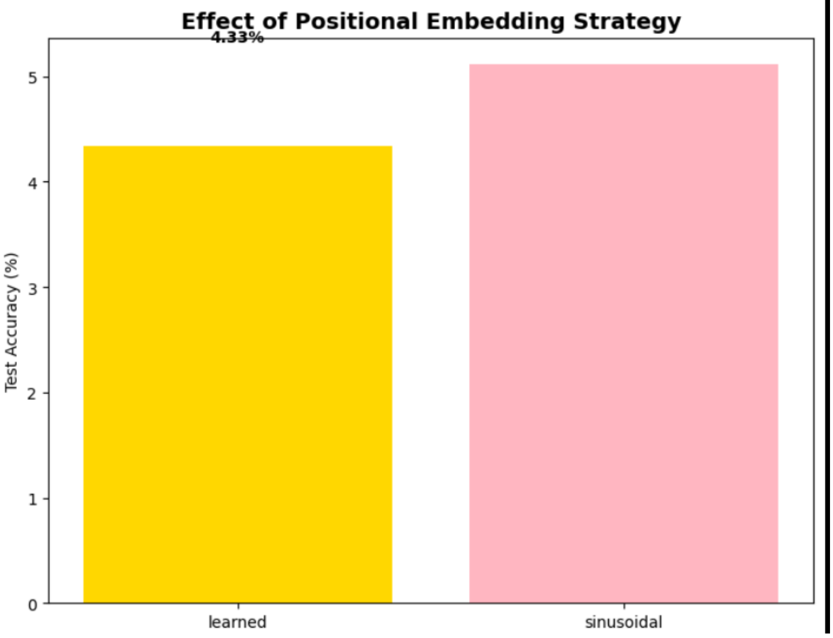

Figure 4: Performance comparison between learned and sinusoidal positional embeddings.

We observed that sinusoidal positional embeddings slightly outperformed learned embeddings (5.1% vs. 4.3%), indicating that fixed positional encoding helps stabilize learning under limited data conditions.

3.5 Model Efficiency Analysis

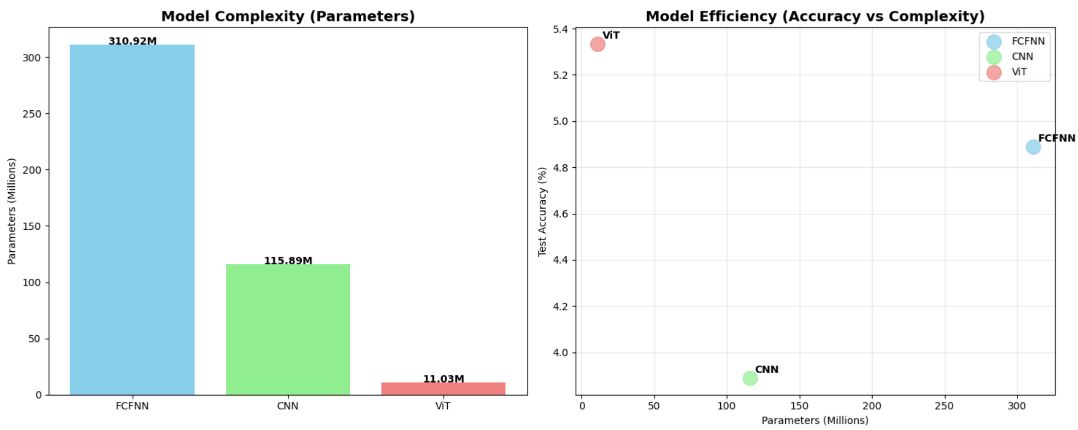

Figure 5: Model complexity (left) and accuracy vs. complexity trade-off (right).

As shown in Figure 5, the ViT exhibits the highest parameter efficiency. Although it has far fewer parameters, it achieves higher accuracy compared to CNN and FCFNN. The FCFNN has the highest parameter count with modest accuracy, highlighting its inefficiency for image-based tasks.

4 Key Findings

4.1 Performance Hierarchy

- The **Vision Transformer (ViT)** achieved the **highest test accuracy (5.33%)** with the **lowest parameter count (11.03M)**, demonstrating strong efficiency.
- The **Fully Connected Feedforward Neural Network (FCFNN)** achieved moderate accuracy (4.89%) but had the **highest parameter count (310.92M)**, indicating parameter inefficiency.
- The **Convolutional Neural Network (CNN)** achieved the **lowest accuracy (3.89%)** with a moderate parameter count (115.89M).

4.2 Optimal ViT Configuration

- **Attention Heads:** 6 heads provided the optimal trade-off between accuracy and complexity.
- **Patch Embedding:** The **overlapping patch embedding** strategy improved accuracy compared to standard patches.
- **Positional Embedding: Sinusoidal embeddings** slightly outperformed learned embeddings in accuracy and stability.

4.3 Trade-offs

- **ViT:** Highly parameter-efficient and effective but requires sufficient training data and careful tuning.
- **CNN:** Provides a good balance between accuracy and computational cost, making it suitable for small and medium-scale tasks.
- **FCFNN:** Simple to implement but performs poorly on high-dimensional image data due to lack of spatial feature extraction.

5 Conclusion

In this study, we compared ViT, CNN, and FCFNN architectures on a **synthetic ImageNet-like dataset** to analyze their relative performance, complexity, and efficiency. We observed that ViT achieved the best accuracy-to-parameter ratio despite limited data, highlighting its scalability and representational power. However, its performance strongly depends on proper configuration and adequate data volume.

Recommendations:

- Use **CNNs** for small datasets and resource-constrained environments.
- Utilize **ViTs** for large-scale datasets where more computation and training time are available.
- Prefer **overlapping patches** and **sinusoidal positional embeddings** for better performance.
- Explore **hybrid CNN–ViT models** to achieve a balance between spatial locality and global context modeling.

Overall, we conclude that the Vision Transformer is a powerful and parameter-efficient architecture whose potential is best realized on larger datasets with proper architectural and hyperparameter tuning.

Code Link:

Here's the github link of the implemented code: https://github.com/mdsajjadhossain25/CSE4261